



**VADANTA INTERNATIONAL SCHOOL**  
**Revision- Trigonometry**

**Question-1**

Prove that  $1 + \frac{\cot^2 \alpha}{1 + \operatorname{cosec} \alpha} = \operatorname{cosec} \alpha$ .

[NCERT Exemplar, CBSE 2019(C) (30/1/1), CBSE 2020 (30/2/1)]

**Question-2**

Prove that  $(\sin \alpha + \cos \alpha)(\tan \alpha + \cot \alpha) = \sec \alpha + \operatorname{cosec} \alpha$ .

[NCERT Exemplar]

**Question-3**

If  $\sin \theta + \cos \theta = \sqrt{3}$ , then prove that  $\tan \theta + \cot \theta = 1$ .

[NCERT Exemplar, CBSE 2020(30/1/1)]

**Question-4**

If  $1 + \sin^2 \theta = 3 \sin \theta \cos \theta$ , prove that  $\tan \theta = 1$  or  $\frac{1}{2}$ .

[CBSE 2020 (30/2/2)]

**Question-5**

Prove that :  $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$

[CBSE 2019 (30/1/2)]

**Question-6**

If  $\sin \theta + \cos \theta = p$  and  $\sec \theta + \operatorname{cosec} \theta = q$ , show that  $q(p^2 - 1) = 2p$ .

[CBSE 2020 (30/3/1)]

**Question-7**

Prove that:  $\frac{1 + \cos \theta - \sin^2 \theta}{\sin \theta (1 + \cos \theta)} = \cot \theta$

**Question-8**

Prove that:  $(\sin \theta + 1 + \cos \theta)(\sin \theta - 1 + \cos \theta) \cdot \sec \theta \operatorname{cosec} \theta = 2$

[CBSE 2019 (30/4/2)]

### Question-9

Prove that:  $\frac{\tan^3 \theta}{1 + \tan^2 \theta} + \frac{\cot^3 \theta}{1 + \cot^2 \theta} = \sec \theta \operatorname{cosec} \theta - 2 \sin \theta \cos \theta$  [CBSE 2019 (30/3/3)]

### Question-10

Prove that :  $\frac{\tan^2 A}{\tan^2 A - 1} + \frac{\operatorname{cosec}^2 A}{\sec^2 A - \operatorname{cosec}^2 A} = \frac{1}{1 - 2 \cos^2 A}$  [CBSE 2019 (30/5/1)]

### Question-11

Prove that:  $(\sin \theta + \sec \theta)^2 + (\cos \theta + \operatorname{cosec} \theta)^2 = (1 + \sec \theta \operatorname{cosec} \theta)^2$

### Question-12

Prove that :  $\frac{(1 + \cot \theta + \tan \theta)(\sin \theta - \cos \theta)}{(\sec^3 \theta - \operatorname{cosec}^3 \theta)} = \sin^2 \theta \cos^2 \theta$  [CBSE 2019 (30/5/1)]

### Question-13

If  $\tan \theta + \sin \theta = m$  and  $\tan \theta - \sin \theta = n$ , show that  $(m^2 - n^2) = 4\sqrt{mn}$ .  
[Competency Based Question]

### Question-14

If  $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$  and  $x \sin \theta = y \cos \theta$ , prove  $x^2 + y^2 = 1$ .  
[Competency Based Question]

### Question-15

Prove that: [CBSE 2019(30/5/1)]  
$$\frac{(1 + \cot \theta + \tan \theta)(\sin \theta - \cos \theta)}{(\sec^3 \theta - \operatorname{cosec}^3 \theta)} = \sin^2 \theta \cos^2 \theta$$

### Question-16

Prove that:  $(1 + \cot \theta + \tan \theta)(\sin \theta - \cos \theta) = \frac{\sec \theta}{\operatorname{cosec}^2 \theta} - \frac{\operatorname{cosec} \theta}{\sec^2 \theta}$

### Question-17

Prove that:

[CBSE 2019 (30/4/3)]

$$2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta) + 1 = 0$$